

Early Intratracheal Budesonide to Reduce Bronchopulmonary Dysplasia in Extremely Preterm Infants

The Budesonide in Babies (BiB) Randomized Clinical Trial

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IMPORTANCE Extremely preterm infants are at high risk for bronchopulmonary dysplasia (BPD) and death. Multiple small randomized clinical trials showed that a combination of budesonide with surfactant compared with surfactant alone reduced BPD or death.

OBJECTIVE To determine if early intratracheal administration of a combination of budesonide (0.25 mg/kg) mixed with surfactant, compared with surfactant alone, reduces physiologic BPD or death by 36 weeks' postmenstrual age in extremely preterm infants.

DESIGN, SETTING, AND PARTICIPANTS This double-masked randomized clinical trial was conducted from April 2021 to June 2024 in the 17 centers of the United States Neonatal Research Network. Infants 22 to 28 weeks' gestation or 401 to 1000 g birth weight were enrolled after clinical decision to give surfactant, with the first dose of surfactant being study drug (prior surfactant was an exclusion criterion).

INTERVENTIONS Infants were randomly allocated 1:1 to receive 1 to 2 doses of budesonide + surfactant (poractant alfa) or surfactant alone via endotracheal tube within 50 hours of birth.

MAIN OUTCOMES AND MEASURES The primary outcome was physiologic BPD or death by 36 weeks' postmenstrual age. There were 5 prespecified secondary outcomes and multiple prespecified exploratory and safety outcomes.

RESULTS The trial was stopped with 641 infants enrolled (55.3% of 1160 planned; mean birth weight, 810 g [SD, 256 g]; gestational age, 25.9 weeks [SD, 1.9 weeks]), because interim analysis at 50% enrollment reached the prespecified futility threshold. The incidence of BPD or death was 68.5% in the budesonide + surfactant group and 67.9% in the surfactant-alone group (adjusted relative risk [RR], 1.00 [95% CI, 0.90-1.11]). No differences were noted in mortality (15.3% vs 13.2%; adjusted RR, 1.13 [95% CI, 0.78-1.64]) or BPD among survivors to 36 weeks' postmenstrual age (62.9% vs 63.0%; adjusted RR, 0.99 [95% CI, 0.87-1.12]). More infants who received budesonide + surfactant compared with surfactant alone had hyperglycemia (66.7% vs 49.8%; adjusted RR, 1.33 [95% CI, 1.17-1.51]).

CONCLUSIONS AND RELEVANCE In this large multicenter trial, the combination of budesonide with surfactant did not reduce the risk of BPD or death at 36 weeks' postmenstrual age in extremely preterm infants.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: [NCT04545866](https://clinicaltrials.gov/ct2/show/study/NCT04545866)

JAMA. 2025;334(16):1452-1462. doi:10.1001/jama.2025.16450
Published online September 30, 2025.

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Infants born extremely preterm (<29 weeks' gestation) or with birth weight 1000 g or less are at high risk of mortality and morbidity.^{1,2} Bronchopulmonary dysplasia (BPD), the most common serious morbidity in such extremely preterm infants,² is associated with long-term adverse respiratory³ and neurodevelopmental outcomes.^{4,5} BPD is defined by the magnitude of respiratory support and supplemental oxygen postnatally at 36 weeks' postmenstrual age⁶⁻⁹ and characterized pathologically by impairment of lung development.¹⁰ While survival of extremely preterm infants has improved in recent decades, the incidence of BPD has stayed constant or increased.^{2,11,12}

Inflammation due to hyperoxia, oxidative stress, ventilation-associated lung injury, and chorioamnionitis likely contributes to BPD.¹³ Systemic corticosteroids administered soon after birth (age <7 days) reduce the combined outcome of BPD or death (BPD/death) before 36 weeks' postmenstrual age but also increase gastrointestinal perforation and cerebral palsy.¹⁴ Direct intratracheal instillation of the corticosteroid budesonide mixed with surfactant reduces lung injury in preterm animal models.¹⁵⁻¹⁷ In preterm infants, intratracheal instillation of budesonide mixed with surfactant reduced BPD/death in small randomized clinical trials (RCTs)^{18,19} and in meta-analyses including these trials.²⁰⁻²² This evidence led to the design and implementation of 2 large multicenter trials conducted nearly simultaneously to evaluate the benefits and risks of combining budesonide with surfactant in extremely preterm infants—the Preventing Lung Disease Using Surfactant + Steroid (PLUSS)²³ and the Budesonide in Babies (BiB) trials. The PLUSS trial (conducted in Australia, New Zealand, Canada, and Singapore) found no difference in survival free of BPD.²³ The BiB trial reported here was conducted in the United States to test the hypothesis that intratracheal administration of budesonide with surfactant, vs surfactant alone, would reduce the frequency of physiologic BPD/death by 36 weeks' postmenstrual age in extremely preterm infants.

Methods

Trial Design and Oversight

The BiB trial was a multicenter, double-masked RCT with 1:1 parallel allocation to budesonide with surfactant (poractant alfa) or surfactant alone (protocol available as [Supplement 1](#); Manual of Operations available as [Supplement 2](#)). BiB was conducted in the Eunice Kennedy Shriver National Institute of Child Health and Human Development Neonatal Research Network under an Investigational New Drug application (142279) approved by the US Food and Drug Administration (FDA). The trial was approved by institutional review boards at each center. RTI International was the data coordinating center (DCC). The independent NRN data and safety monitoring board, in collaboration with the DCC, conducted interim analyses after the first 40 infants reached the primary outcome and after 25% and 50% of patients reached this milestone to assess safety, efficacy, and/or futility (statistical analysis plan available as [Supplement 3](#)). This report follows the CONSORT (Consolidated Standards of Reporting Trials) guideline.²⁴

Key Points

Question Does intratracheal administration of budesonide mixed with surfactant, compared with surfactant alone, reduce bronchopulmonary dysplasia or death by 36 weeks' postmenstrual age in preterm infants less than 29 weeks' gestation?

Findings In this multicenter randomized trial, after recruiting 641 infants, recruitment was stopped early when prespecified futility criteria were satisfied. There was no difference in bronchopulmonary dysplasia or death between infants receiving budesonide with surfactant (68.5%) vs those receiving surfactant alone (67.9%; adjusted relative risk 1.00 [95% CI, 0.90-1.11]).

Meaning In extremely preterm infants, mixing budesonide with surfactant did not reduce bronchopulmonary dysplasia or death.

Participants

Infants 22 weeks 0 days' to 28 weeks 6 days' gestational age or 401 to 1000 g birth weight were eligible after a clinical decision was made at 48 hours' or less postnatal age to administer surfactant. Infants were excluded if they were judged unlikely to survive or had a decision to redirect or limit support, or there was (1) use of surfactant or systemic steroids before enrollment; (2) maternal exposure to indomethacin within 24 hours of delivery or infant exposure to indomethacin before enrollment; (3) intent to use indomethacin within 7 days of the last dose of study drug; (4) serious chromosomal abnormality or major malformation, known congenital infection, permanent neuromuscular condition affecting respiration; or (5) enrollment in a conflicting clinical trial.

Randomization

Written parental informed consent was obtained before or after birth. Study drug was administered within 50 hours after birth. Infants were stratified by study site and by gestational age (<26 or ≥26 weeks). Multiple births were randomized independently. Step-forward randomization was implemented to enable expeditious delivery room randomization, adopting "use next" drug kits with a block urn design.^{25,26} Based on a DCC-generated randomization schedule, drug kits of budesonide respules or sham (empty respules) were randomized, assembled, masked, and labeled by the investigational drug/research pharmacy at each site, then stored in a location that permitted rapid access by the respiratory therapist or designee. Drug kits were added to the queue such that there were 3 or more sets of kits available for each gestational stratum at any given time. Two drug kits were prepared for each randomization number in case a second dose was needed. Infants were considered randomized when the masked drug kit was opened immediately before treatment administration. Infants of multiple gestations were randomized separately. Infants were randomized 1:1 to either budesonide (0.25 mg/kg; 1 mL/kg) mixed with surfactant (poractant alfa [2.5 mL/kg]) (intervention) or to surfactant (2.5 mL/kg) alone (active comparator).

Interventions

The study drug was administered through an endotracheal tube. Minimally invasive or less invasive surfactant administration

was not used for the trial because it is not an FDA-approved method of surfactant administration. A maximum of 2 doses of study drug could be administered before 50 hours postnatal age, with the second dose containing budesonide (1 mL/kg) and surfactant (1.25 mL/kg). The study drug was administered by an unmasked (due to need for mixing of study drug) respiratory therapist or qualified designee out of sight of other staff, so other caregivers and researchers remained masked to treatment allocation. Infants receiving more than 2 surfactant doses within 50 hours after birth could receive additional doses of open-label surfactant at clinician discretion.

Administration of systemic steroids within less than 7 days of last dose of study drug was discouraged and reserved at clinician discretion for infants with fluid-resistant/vasopressor-resistant hypotension or evidence of hyponatremia with low serum cortisol concentration (per local definitions). Other clinical practices were per usual clinical protocol and clinician preference.

Outcomes

The primary outcome was the composite of physiologic BPD^{6,27,28} or death by 36 weeks' postmenstrual age. Physiologic BPD was defined as receiving mechanical ventilation or continuous positive airway pressure, oxygen by hood at fraction of inspired oxygen (FIO_2) greater than 0.3, oxygen by nasal cannula at an effective FIO_2 greater than 0.3 (calculated using conversion tables), or failing oxygen reduction challenge from FIO_2 less than 0.3 by hood or effective FIO_2 less than 0.3 by cannula (eAppendix in Manual of Operations and Generic Database Manual [Supplement 2]; adjudication was not necessary for this validated definition). Infants discharged home or transferred out of network before 36 weeks' postmenstrual age were assessed for physiologic BPD based on the level of respiratory support at discharge or transfer.

Secondary outcomes assessed at 36 weeks' postmenstrual age included death; physiologic BPD; BPD severity (ordinal)⁷; grade 3 BPD⁷; and postnatal steroid use from 7 days after last dose of study drug through 36 weeks' postmenstrual age.

Multiple exploratory outcomes were also evaluated (Protocol in Supplement 1): BPD severity by the National Institutes of Health Workshop definition⁹; days receiving mechanical ventilation by 28 postnatal days; days receiving invasive mechanical ventilation by 36 weeks' postmenstrual age; intubation after 50 postnatal hours and within the first 28 days; intubation after 50 hours and before 36 weeks' postmenstrual age; and repeat administration of surfactant doses after 50 hours and within 7 days of last dose of study drug. Additional outcomes analyzed include common clinical outcomes assessed through 120 postnatal days and standardized growth metrics at 36 weeks' postmenstrual age. Analyses of survival without physiologic BPD at 36 weeks' postmenstrual age were conducted within subgroups defined by baseline characteristics.

Prespecified adverse events (defined in the protocol in Supplement 1 and Manual of Operations in Supplement 2) considered likely to be associated with study intervention were monitored within 7 days of last dose of study drug. These included hyperglycemia, hypertension, prolonged hypoxemia with bradycardia, endotracheal tube blockage, and any

other adverse event deemed moderate or worse in severity. Moderate hyperglycemia was defined as blood glucose level 120 mg/dL (6.66 mmol/L) to 180 mg/dL (9.99 mmol/L) on 2 consecutive determinations at least 6 hours apart or more than 2 determinations within a 24-hour period with no in-between normal range values. Severe hyperglycemia was defined as blood glucose level greater than 180 mg/dL. Spontaneous intestinal perforation and periventricular leukomalacia within 30 days of final drug administration were also monitored as prespecified adverse events. Several risks of prematurity were monitored within 7 days of last study drug (hypotension, pulmonary air leak, culture-positive sepsis, and intracranial hemorrhage).

Sample Size Estimation and Statistical Analysis

A detailed statistical analysis plan for the BiB trial is available in Supplement 3.

Sample size calculations were based on the primary outcome. Based on the 20% or greater absolute risk reduction observed by prior RCTs^{18,19} and the observed incidence of BPD or death of 58% at participating sites, BiB was designed to detect an absolute risk reduction of 10% (from 58% to 48%), corresponding to a relative risk (RR) of 0.83. Simulation studies for the primary analysis method indicated that a sample size of 550 per group (1100 total) would attain 90% power for an overall type I error of .05. Accounting for 5% attrition for the primary outcome, the planned sample size was 1160 infants.

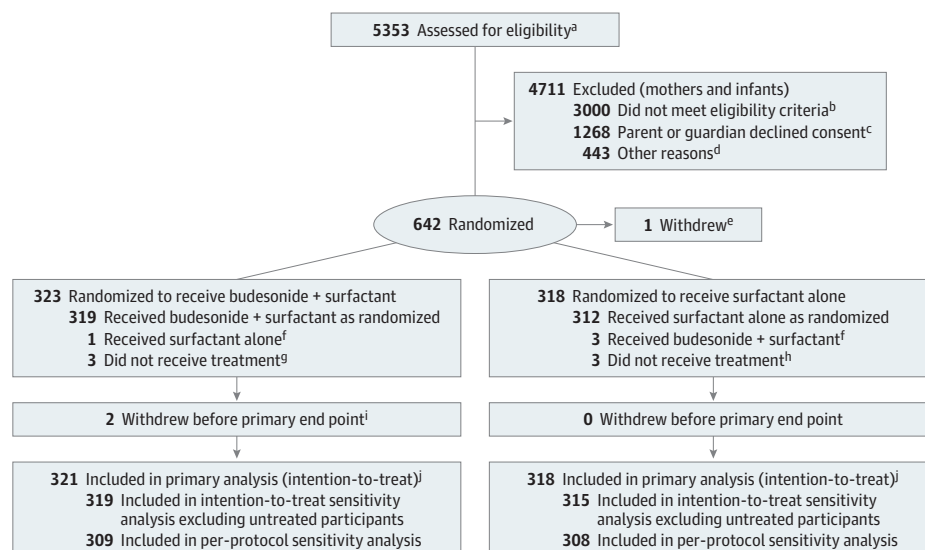
The primary analysis compared the proportion of infants who died or developed BPD at 36 weeks' postmenstrual age using a robust Poisson regression model.^{29,30} Randomization stratification factors of center and gestational age (<26 or ≥26 weeks) were included as fixed explanatory variables in the model. The primary outcome analysis was performed at $\alpha = .049$ to adjust for multiplicity of interim efficacy analyses and preserve an overall type I error rate of $\alpha = .05$. Analyses of the primary, secondary and exploratory efficacy outcomes were based on intention-to-treat (ITT), unless otherwise specified. In contrast, safety, clinical, and growth outcomes were analyzed based on actual treatment received ("as-treated" population). Secondary and exploratory dichotomous outcomes were analyzed with the robust Poisson methods described for the primary outcome. For models with convergence or fit issues, the RR was estimated by the Mantel-Haenszel common odds ratio, stratified by the cross-classification of center and gestational age strata, or by the crude odds ratio (unadjusted) and Fisher exact test. Continuous outcomes estimated adjusted mean difference via linear regression; count outcomes were analyzed as continuous outcomes using robust variance estimation. Comparisons of secondary outcomes between study groups should be considered exploratory.

Results

Trial Population

Patients were randomized from April 2021 to June 2024. When 641 infants (55.3% of 1160 planned) had been enrolled from

Figure 1. Study Flow for a Trial of Early Intratracheal Budesonide to Reduce Bronchopulmonary Dysplasia in Extremely Preterm Infants



^aAssessments of eligibility include antenatal screenings of mothers (n = 4042) and postnatal screenings of infants (n = 1311). Families who declined consent antenatally or who consent antenatally but delivered out of gestational age window were each captured once, regardless of single or multiple gestation.

^bReasons for ineligibility include prior use of surfactant (1356), no clinical decision to give surfactant (811), gestational age and/or birth weight outside eligibility window (387), prior indomethacin use (224), terminally ill or a decision to redirect support (193), serious chromosomal abnormality or neuromuscular condition (138), more than 48 hours' postnatal age at time of randomization (59), enrolled in a conflicting trial (58), stillbirth (47), prior use of systemic steroids (20), and known congenital infection (15). Multiple reasons for ineligibility may be provided.

^cA total of 815 parents declined consent antenatally and prior to assessment of infant eligibility; 453 parents declined consent postnatally, of which 399 infants were eligible and 54 had not been assessed for eligibility.

^dOther reasons not enrolled include passive nonconsent (eg, parent/guardian no decision, not approached, or unavailable), physician decision not to consent and/or enroll, or logistic barriers to the consent and/or enrollment process (eg, staff availability, drug kit availability, urgent delivery, parent/guardian did not speak English, early study closure).

^eOne infant was randomized and treated in error after the parent/guardian declined consent. The infant was withdrawn from the trial on discovery of the protocol violation. At the parent's request, the infant's data have been fully redacted. No data are available for reporting or analysis (including treatment

allocation), and this participant is excluded from the intention-to-treat population.

^fParticipants who received budesonide with surfactant for either or both doses of study drug are considered to be treated under the budesonide + surfactant group.

^gOne infant died prior to treatment; 1 became too medically unstable to participate; 1 was ineligible and randomized in error.

^hTwo infants improved and did not require surfactant; one was ineligible and randomized in error.

ⁱIn addition to the withdrawn infant (see footnote "e"), 2 randomized infants withdrew prior to primary end point assessment. One infant was randomized and treated without consent (parent/guardian declined consent to continue). The other infant was ineligible and randomized in error (not treated); effectively withdrawn since they were also ineligible for the Generic Data Base data collection.

^jThe primary analysis was performed for the intention-to-treat analysis population, and the model excluded participants missing the primary end point (2). This model was repeated for 2 analysis subpopulations as a planned sensitivity analysis. The intention-to-treat analysis excluding untreated participants also excluded participants who were not treated (5). The per-protocol analysis also excluded participants who either were not treated (5) or had a major treatment protocol violation (5 ineligible enrollments; 4 treated under the wrong group or both groups; 3 received more than 2.4 times the intended dose; 2 received less than one-half of the intended dose; 1 treated after the dosing window; and 2 other dosing administration violation).

17 centers, the data and safety monitoring board in June 2024 recommended study termination at 50% enrollment based on having met the prespecified futility criterion (80% CI for conditional power falling below .50)³¹⁻³³ (statistical analysis plan in Supplement 3). A total of 5353 antenatally screened mothers (n = 4042) and postnatally screened infants (n = 1311) were assessed for eligibility (Figure 1). Consent was not obtained in 4042 mothers/infants due to lack of eligibility (n = 1784), consent declined (n = 1417), or other reasons (n = 841). Of 1311 mothers/infants with consent, 670 were subsequently eligible (Figure 1). In total, 641 infants were randomized (637 of the 670 eligible and consented, 1 nonconsented, and 3 ineligible), with 323 randomized to budesonide + surfactant and 318 to surfactant alone. The primary end point at 36 weeks' postmenstrual age was available for all but 2 infants in the in-

tervention group. Additional information about participant study experience is tabulated in Supplement 4.

The maternal and neonatal baseline clinical characteristics were similar between the 2 groups (Table 1). Overall, the mean gestational age was 25.9 weeks (SD, 1.9), and the mean birth weight was 810 g (SD, 256), with 41.8% in the less than 26 weeks stratum, 50.1% male, and 23.0% born following multiple-gestation pregnancies. Nearly all infants (99.1%) were exposed to antenatal steroids, with 89.8% exposed to a full course; in all, 96.1% were exposed to magnesium sulfate and 71.2% to antibiotics within 72 hours of birth. At delivery, 85.1% of infants received positive pressure ventilation and 64.5% received continuous positive airway pressure. In all, 58.5% of infants were intubated at birth, and about one-half those infants (27.6% overall) received their first dose of study drug in the delivery room.

Table 1. Maternal, Labor and Delivery, and Neonatal Baseline Characteristics in the Intention-to-Treat Population

Characteristic	Budesonide + surfactant (n = 323)	Surfactant alone (n = 318)
Maternal		
Age, mean (SD), y	29.1 (6.1)	29.7 (5.8)
Race, self-reported, No. (%)	n = 305	n = 307
American Indian or Alaska Native	5 (1.6)	4 (1.3)
Asian, Native Hawaiian, or Other Pacific Islander	12 (3.9)	8 (2.6)
Black or African American	101 (33.1)	122 (39.7)
White	183 (60.0)	168 (54.7)
More than 1 race	4 (1.3)	5 (1.6)
Hispanic or Latino ethnicity, No./total (%)	50/304 (16.4)	54/308 (17.5)
Education, No. (%)	n = 259	n = 269
<High school diploma	31 (12.0)	27 (10.0)
High school diploma	87 (33.6)	77 (28.6)
Partial college	56 (21.6)	81 (30.1)
College degree or more	85 (32.8)	84 (31.2)
Health insurance, No. (%)	n = 321	n = 318
Private	160 (49.8)	160 (50.3)
Public	149 (46.4)	143 (45.0)
Self-pay/uninsured	12 (3.7)	15 (4.7)
Body mass index, prepregnancy, mean (SD)	30.8 (8.9)	31.4 (8.8)
Hypertensive disorder of pregnancy, No./total (%)	142/321 (44.2)	164/318 (51.6)
Diabetes, prepregnancy, No./total (%)	16/321 (5.0)	29/318 (9.1)
Diabetes, gestational, No./total (%)	22/298 (7.4)	16/296 (5.4)
Chorioamnionitis, clinical, No./total (%)	39/321 (12.1)	38/318 (11.9)
Chorioamnionitis, histologic, No./total (%)	118/300 (39.3)	112/302 (37.1)
Multiple birth, No./total (%)	78/321 (24.3)	69/318 (21.7)
Labor and delivery		
Antenatal steroids, any, No./total (%)	318/321 (99.1)	314/317 (99.1)
Antenatal steroids, full course, No./total (%)	282/319 (88.4)	288/316 (91.1)
Magnesium sulfate, No./total (%)	307/321 (95.6)	306/317 (96.5)
Antibiotics given <72 h prior to delivery, No./total (%)	236/321 (73.5)	219/318 (68.9)
Prolonged rupture of membrane (>18 h), No./total (%)	98/319 (30.7)	89/315 (28.3)
Cesarean delivery, No./total (%)	243/321 (75.7)	236/318 (74.2)
Delayed cord clamping, No./total (%)	146/321 (45.5)	156/317 (49.2)
Infant		
Birth weight, mean (SD), g	808.1 (268.8)	812.3 (242.6)
Gestational age, mean (SD), wk	25.8 (1.9)	25.9 (2.0)
Gestational age <26 wk 7 d, No./total (%)	138/323 (42.7)	130/318 (40.9)
Sex, No./total (%)		
Male	158/321 (49.2)	162/318 (50.9)
Female	163/321 (50.8)	156/318 (49.1)
Small for gestational age (<10th percentile for gestational age and sex), No./total (%)	62/321 (19.3)	50/318 (15.7)
Apgar score, median (IQR)		
1 min	3 (2-5)	4 (2-6)
5 min	7 (5-8)	7 (6-8)

(continued)

Table 1. Maternal, Labor and Delivery, and Neonatal Baseline Characteristics in the Intention-to-Treat Population (continued)

Characteristic	Budesonide + surfactant (n = 323)	Surfactant alone (n = 318)
Delivery room care, No./total (%)		
Positive pressure ventilation	278/320 (86.9)	265/318 (83.3)
Continuous positive airway pressure	202/319 (63.3)	208/317 (65.6)
Chest compression	4/321 (1.2)	4/318 (1.3)
Epinephrine	3/321 (0.9)	4/318 (1.3)
Intubation ^a	191/321 (59.5)	183/318 (57.5)
Type of respiratory support at baseline, No. (%) ^b	n = 232	n = 235
Continuous positive airway pressure	115 (49.6)	119 (50.6)
Conventional mechanical ventilation	67 (28.9)	62 (26.4)
High-frequency ventilation	45 (19.4)	47 (20.0)
Hood	1 (0.4)	2 (0.9)
None	4 (1.7)	5 (2.1)
IMV respiratory support at baseline, No./total (%) ^b	112/232 (48.3)	109/235 (46.4)
FiO ₂ at baseline, median (IQR) ^b	0.40 (0.30-0.60) [n = 228]	0.40 (0.30-0.56) [n = 230]
High FiO ₂ (≥0.5) at baseline, No./total (%) ^b	86/228 (37.7)	81/230 (35.2)
Partial pressure of carbon dioxide at baseline, mean (SD), mm Hg ^b	52.0 (14.2) [n = 164]	50.8 (14.6) [n = 156]
Blood gas pH at baseline, mean (SD) ^b	7.3 (0.1) [n = 164]	7.3 (0.1) [n = 156]
Study drug exposures		
No. of doses, No. (%)		
1	221 (69.1)	194 (61.6)
2	99 (30.9)	121 (38.4)
Administered in delivery room, No./total (%) ^c	90/318 (28.3)	83/308 (26.9)
Postnatal age at exposure, mean (SD), h		
Dose 1	4.07 (7.49) [n = 320]	3.68 (6.78) [n = 315]
Dose 2	23.38 (11.85) [n = 99]	21.76 (10.10) [n = 121]

Abbreviations: FiO₂, fraction of inspired oxygen; IMV, invasive mechanical ventilation.

^a Resuscitation/stabilization at birth with intubation excludes intubations that were performed for suctioning or surfactant administration and then immediately removed.

^b Baseline respiratory and clinical blood gas metrics report the last available data collected prior to study drug initiation. These data collections were optional and have high levels of missingness not at random. Most infants who received study drug within 1 hour of birth do not have baseline collections, and some infants may have no baseline data or baseline data that were captured several hours before treatment initiation.

^c By study design, all surfactant administrations in delivery room were the first study drug intervention.

Primary Outcome

Physiologic BPD or death by 36 weeks' postmenstrual age occurred in 68.5% of infants in the budesonide + surfactant group and 67.9% in the surfactant-alone group (adjusted RR, 1.00 [95% CI, 0.90-1.11]) (Table 2).

Secondary Outcomes

There were no differences between treatment groups in death (15.3% budesonide + surfactant vs 13.2% surfactant alone; adjusted RR, 1.13 [95% CI, 0.78-1.64]) or physiologic BPD (62.9% budesonide + surfactant vs 63.0% surfactant alone; adjusted RR, 0.99 [95% CI, 0.87-1.12]). Other outcomes including BPD severity⁷ at 36 weeks' postmenstrual age; grade 3 BPD⁷ at 36 weeks' postmenstrual age; and postnatal steroid use from 7 days' poststudy drug administration through 36 weeks' postmenstrual age also did not differ between the groups (Table 3).

Exploratory Outcomes

Several predefined exploratory efficacy outcomes were also evaluated. BPD severity by the National Institutes of Health Workshop definition⁹ did not differ between the groups, but there were fewer days receiving mechanical ventilation before 28 days in the budesonide + surfactant group (median, 7 [IQR, 1-26] days vs 10 [IQR, 2-28] days; mean difference, -1.49

[95% CI, -2.97 to -0.01]). There were fewer infants exposed to invasive ventilation after the treatment window (continued on, or new intubation) in the budesonide + surfactant group compared with the surfactant-alone group before postnatal day 28 (61.5% vs 68.2%; adjusted RR, 0.87 [95% CI, 0.78-0.96]) and before 36 weeks' postmenstrual age (64.8% vs 71.5%; adjusted RR, 0.87 [95% CI, 0.79-0.96]). Open-label surfactant doses after the treatment window and within 7 days after the last dose of the study drug did not differ between the groups (Table 3).

Safety Outcomes

The probability of experiencing at least 1 adverse event differed between the groups (75.4% budesonide + surfactant vs 64.5% surfactant alone; adjusted RR, 1.16 [95% CI, 1.05-1.28]) (Supplement 4), primarily due to an increase in moderate hyperglycemia without a difference in any serious adverse events.

Clinical and Growth Outcomes

Clinically diagnosed patent ductus arteriosus (PDA) was less common in the budesonide + surfactant compared with the surfactant-alone group (49.8% vs 56.8%; adjusted RR, 0.86 [95% CI, 0.75-0.99]), but no differences were observed for PDA managed with medical therapy, PDA managed with surgery or

Table 2. Primary Efficacy Outcome and Its Components^a

Efficacy outcome	No./total (%)		Risk difference (95% CI)	Relative risk (95% CI)
	Budesonide + surfactant (n = 323)	Surfactant alone (n = 318)		
Primary outcome ^b				
Physiologic BPD or death before 36 wk PMA				
ITT	220/321 (68.5)	216/318 (67.9)	0.02 (−6.81 to 6.85)	1.00 (0.90 to 1.11)
ITT excluding untreated participants	218/319 (68.3)	215/315 (68.3)	−0.31 (−7.18 to 6.56)	1.00 (0.90 to 1.10)
Per-protocol	212/309 (68.6)	210/308 (68.2)	−0.14 (−7.11 to 6.83)	1.00 (0.90 to 1.11)
Primary components, ITT				
Death before 36 wk PMA	49/321 (15.3)	42/318 (13.2)	1.73 (−3.56 to 7.01)	1.13 (0.78 to 1.64)
Physiologic BPD at 36 wk PMA ^c	171/272 (62.9)	174/276 (63.0)	−0.73 (−8.38 to 6.93)	0.99 (0.87 to 1.12)

Abbreviations: BPD, bronchopulmonary dysplasia; ITT, intention-to-treat, postmenstrual age; PMA, postmenstrual age.

^a Modeled estimates compare the budesonide + surfactant group with the surfactant-alone group, adjusting for gestational age strata and pooled center (7-level). Risk differences and CIs (reported as %) were estimated with robust generalized linear regression (normal distribution with identity link); relative risks and CIs were estimated by robust Poisson regression. Treatment group represents randomized treatment assignment for all analysis populations in this table.

^b Primary outcome models were each adjusted for multiplicity from sequential testing (performed at $\alpha = .049$). Multiplicity adjustment was not applied for multiple testing of sensitivity analyses. All other models were performed at $\alpha = .05$ for descriptive purposes only.

^c The physiologic definition of BPD was used for the primary outcome (detailed definition in Supplement). Infants were evaluated to see if they were eligible for the oxygen reduction challenge if on day of 36 weeks' PMA, they used nasal cannula with or without supplemental oxygen, or oxygen hood/oxygen delivered into incubator. The effective oxygen concentration (effective FiO_2) was calculated based on infant's weight, flow, and oxygen concentration using conversion tables. Infants were considered eligible for the oxygen reduction challenge if they met 1 of the following: (1) effective oxygen less than 27% and majority of saturations 90% or greater in prior 24 hours; (2) effective oxygen 27% to 30% and majority of saturations 96% or greater; or (3) room air by nasal cannula.

cardiac catheterization, or the composites of death and PDA outcome (Supplement 4). No other clinical or growth differences were observed between treatment groups.

Sensitivity and Subgroup Analyses

The primary efficacy outcome was reassessed for the ITT population excluding untreated participants (all randomized patients who received at least 1 dose of study drug) ($n = 635$) and the per-protocol population ($n = 617$). Both the ITT analysis excluding untreated participants and the per-protocol analysis excluded 6 randomized infants who never initiated treatment; the per-protocol analysis also excluded 18 infants with a major protocol violation (Supplement 4). No differences were detected between groups for these sensitivity analyses (Table 2).

No significant differences in survival without physiologic BPD at 36 weeks' postmenstrual age by treatment group were observed for any subgroups defined by baseline characteristics (Figure 2).

Discussion

In this multicenter RCT, administration of intratracheal budesonide mixed with surfactant did not reduce BPD or death in extremely preterm infants compared with surfactant alone. The intervention was associated with fewer days receiving mechanical ventilation by postnatal day 28 as well as an increased incidence of hyperglycemia. The results of this study are consistent with the findings from the PLUSS trial,²³ which also found that the combination of budesonide and surfactant did not significantly improve survival free of BPD in extremely preterm infants.

This trial provides a rigorous evaluation of the intervention in a large and diverse high-risk population of extremely preterm infants, using strict inclusion and exclusion criteria, double-masking, well-defined prespecified outcomes, and detailed analyses. These extremely preterm infants received high-risk obstetric care (99% exposed to antenatal steroids; 96% to magnesium sulfate; 75% to cesarean delivery) and had high survival (86% survival to 36 weeks' postmenstrual age). Infants also received the study drug through an endotracheal tube, ensuring that study drug was administered into the lungs. Study drug was given as the first dose of surfactant, thereby ensuring early use of the intervention. Infants who had already received surfactant were excluded, avoiding prior exposure that could blur potential efficacy or harm signals.

Neither this trial nor a prior multinational trial²³ identified a benefit of budesonide combined with surfactant on survival free of BPD among extremely preterm infants. The populations in both trials were diverse—in the current trial, 36% of infants were born to Black mothers and 17% to Hispanic mothers, while the multinational trial included infants born to mothers of primarily Asian, Caucasian, Maori, and Pacific Islander origins.

The key novel contribution of the current trial is that administration of the study intervention with the first dose of surfactant more closely parallels the way budesonide would be used in real-world practice, in contrast to the multinational trial, where 57% of infants had received prior surfactant. Another difference is that about 14% of the infants in the prior trial received surfactant via a thin catheter, which may be safe and associated with reductions in BPD.³⁴ However, this was not permitted in the current trial because surfactant administration via endotracheal tube is the only method currently approved

Table 3. Secondary and Exploratory Efficacy Outcomes in the Intention-to-Treat Population

Outcome	Budesonide + surfactant (n = 323)	Surfactant alone (n = 318)	Risk difference or mean difference (95% CI) ^a	Relative risk or odds ratio (95% CI) ^a
Secondary outcomes				
BPD severity at 36 wk PMA, pragmatic, No. (%) ^b	269	270		
No BPD	70 (26.0)	66 (24.4)		1 [Reference]
Grade 1 (mild)	72 (26.8)	78 (28.9)		OR: 0.87 (0.54 to 1.41)
Grade 2 (moderate)	99 (36.8)	100 (37.0)		OR: 0.88 (0.55 to 1.42)
Grade 3 (severe)	28 (10.4)	26 (9.6)		OR: 0.99 (0.51 to 1.89)
Grade 3 (severe) BPD at 36 wk PMA, pragmatic, No./total (%) ^b	28/269 (10.4)	26/270 (9.6)	RD: 0.69 (−4.34 to 5.71)	RR: 1.07 (0.65 to 1.77)
Use of additional postnatal steroids between 7 d PLD and 36 wk PMA, No./total (%) ^c	102/318 (32.1)	109/311 (35.0)	RD: −3.84 (−10.45 to 2.77)	RR: 0.89 (0.73 to 1.09)
Exploratory outcomes				
BPD severity at 36 wk PMA, NIH consensus definition, No. (%) ^d	270	270		
No BPD	29 (10.7)	27 (10.0)		1 [Reference]
Grade 1 (mild)	70 (25.9)	57 (21.1)		OR: 1.13 (0.59 to 2.17)
Grade 2 (moderate)	32 (11.9)	45 (16.7)		OR: 0.63 (0.30 to 1.34)
Grade 3 (severe)	139 (51.5)	141 (52.2)		OR: 0.87 (0.47 to 1.63)
No. of days receiving IMV before 28 d postnatal age ^e	(n = 283)	(n = 278)		
Median (IQR)	7 (1–26)	10 (2–28)	MD: −1.49 (−2.97 to −0.01)	NA
No. of days receiving IMV before 36 wk PMA ^e	(n = 262)	(n = 265)		
Median (IQR)	10 (1–36)	13 (2–37)	MD: −2.04 (−5.32 to 1.23)	NA
Intubated after treatment window and before 28 d postnatal age, No./total (%) ^f	192/312 (61.5)	206/302 (68.2)	RD: −9.09 (−15.48 to −2.69)	RR: 0.87 (0.78 to 0.96)
Intubated after treatment window and before 36 wk PMA, No./total (%) ^f	195/301 (64.8)	211/295 (71.5)	RD: −9.12 (−15.50 to −2.73)	RR: 0.87 (0.79 to 0.96)
Additional open-label surfactant after treatment window and by 7 d after last dose of study drug, No./total (%) ^{g,h}	13/312 (4.2)	18/299 (6.0)	RD: −1.87 (−5.29 to 1.55)	RR: 0.69 (0.33 to 1.46)

Abbreviations: BPD, bronchopulmonary dysplasia; IMV, invasive mechanical ventilation; MD, mean difference; NA, not applicable; OR, odds ratio; RD, risk difference; RR, relative risk; PMA, postmenstrual age.

^a Binary outcomes report RDs (as %) and RRs; ordinal outcomes report ORs; count outcomes report least-squares MDs. Risk differences and least-square MDs were estimated by robust generalized linear models (normal distribution with identity link); RRs were estimated by robust Poisson regression; and ORs were estimated by logistic regression (generalized logit). RDs are not shown for BPD severity because reduced sample sizes lead to very wide CIs for pairwise comparisons. All models compared the budesonide + surfactant group with the surfactant-alone group, adjusting for gestational age strata and pooled center. All analyses were performed at the $\alpha = .05$ significance level for descriptive purposes only.

^b The pragmatic definition of BPD severity grades was defined by Jensen et al.⁷ No BPD is defined as no support at 36 weeks' PMA; grade 1 as oxygen delivered via nasal cannula at less than 2 L/min; grade 2 as oxygen delivered via nasal cannula at more than 2 L/min or noninvasive positive airway pressure; and grade 3 as invasive mechanical ventilation.

^c "Additional postnatal steroids" specifically refer to steroids used to treat evolving BPD or chronic lung disease. This outcome is limited to those who were alive and in hospital at the end of the 7 days after last-dose safety monitoring window.

^d The National Institutes of Health consensus definition of BPD severity grades was defined by Jobe and Bancalari.⁹ The scoring algorithm considers an infant's gestational age, duration of respiratory support through 28 days' postnatal age, and the level of respiratory support at 36 weeks' PMA.

^e IMV includes high-frequency ventilation and conventional ventilation; nasal cannula respiratory support is not considered invasive per this protocol. IMV outcomes are limited to infants who were alive and in hospital at the specified analysis end point (28 days' postnatal age or 36 weeks' postmenstrual age).

^f Intubations after the treatment window include new or continued intubations after 50 hours' postnatal age. Intubation outcomes are limited to infants who were alive and in hospital at 50 hours' postnatal age.

^g Relative risk was estimated by Mantel-Haenszel methods, approximated by the common OR, due to convergence issues with the robust Poisson model. The analysis was stratified by the cross-classifications of gestational age strata and pooled center.

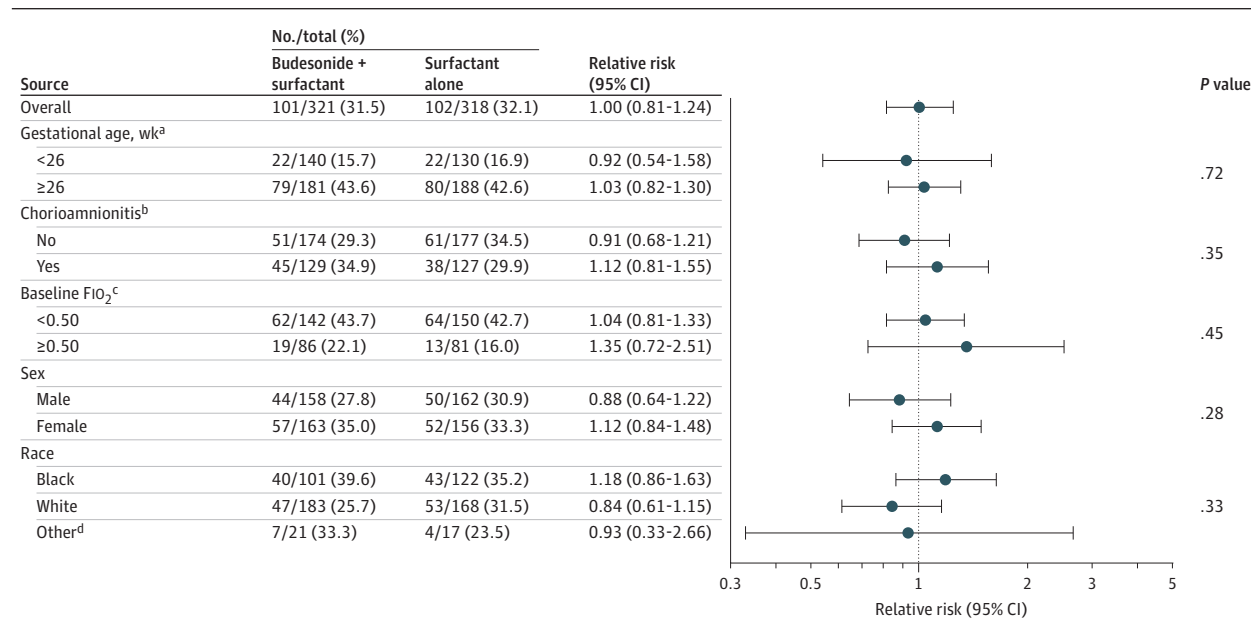
^h Additional surfactant administrations do not include any open-label administrations within the treatment window (50 hours' postnatal age). This outcome is limited to infants who were alive and in hospital at 50 hours' postnatal age.

by the FDA. Despite such differences, the similar results of these 2 multicenter trials indicate that these differences are unlikely to influence the utility of the intervention.

Benefits of budesonide with surfactant were noted in trials by Yeh et al^{18,19} that enrolled infants who were generally more

mature at birth, less likely to be exposed to chorioamnionitis or antenatal steroids, met specific FiO_2 threshold requirements, and had lower incidence of BPD/death in controls compared with the 2 multicenter trials. This study did not observe a difference in the post hoc subgroup analysis to assess

Figure 2. Survival Without Physiologic Bronchopulmonary Dysplasia at 36 Weeks' Postmenstrual Age by Subgroup in the Intention-to-Treat Population



Subgroup analyses considered relative risk of a favorable outcome (alive and without physiologic bronchopulmonary dysplasia) at 36 weeks' postmenstrual age, comparing the budesonide + surfactant group with the surfactant-alone group. Relative risks and 95% CIs were estimated using robust Poisson regression. The gestational age subgroup models adjusted for pooled center; all other models adjusted for gestational age strata and pooled center. All analyses were performed at the $\alpha = .05$ significance level for descriptive purposes only. FIO₂ indicates fraction of inspired oxygen.

^aReflects actual gestational age (best clinical estimate), regardless of randomization strata.

^bHistological or clinical.

^cRefers to the last known level of respiratory support prior to treatment initiation. The FIO₂ subgroups exclude 182 participants from the intention-to-treat population who do not have baseline respiratory data. Most of the excluded participants (138, 76%) initiated treatment within 30 minutes of birth.

^dIncludes self-reported American Indian or Alaska Native, Asian, Native Hawaiian or Other Pacific Islander, or more than 1 race.

the influence of baseline FIO₂ on intervention effect. It is possible that differences in gestational age, exposure to antenatal steroids, clinical practices,³⁵ genetic predispositions,³⁶ and other factors led to differences in effects of interventions and frequency of BPD. It is unlikely that higher doses of budesonide are necessary, because even one-tenth the dose of budesonide (0.025 mg/kg) appears effective for lung-targeted anti-inflammatory action.³⁷

Limitations

This study's limitations include, first, that early termination of the trial resulted in a sample size smaller than planned, reducing power for secondary outcomes and subgroup analyses.

Second, the results from BiB may be generalizable only to extremely preterm infants with similar characteristics (high exposure to antenatal steroids, cesarean delivery, and maternal hypertension), such as those seen in the *Eunice Kennedy Shriver* National Institute of Child Health and Human Development Neonatal Research Network² and the Vermont Oxford Network.^{38,39}

Conclusions

In extremely preterm infants, intratracheal budesonide mixed with surfactant did not reduce BPD or death, or the components of this outcome, by 36 weeks' postmenstrual age.

ARTICLE INFORMATION

Accepted for Publication: August 19, 2025.

Published Online: September 30, 2025.
doi:10.1001/jama.2025.16450

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Conflict of Interest Disclosures: Dr Ambalavanan reported serving on the data and safety monitoring board for OakHill Bio; serving as medical advisor to ResBiotic; and owning stock in AlveolusBio outside the submitted work. Dr D'Angio reported receiving grants from NIH outside the submitted work.

Dr Cotten reported serving on the advisory board for ReALTA Life Sciences outside the submitted work. Dr Laughon reported receiving grants from NIH outside the submitted work. Dr Hibbs reported receiving grants from the National Heart, Lung, and Blood Institute outside the submitted work. Dr DeMauro reported receiving grants from NIH outside the submitted work. Dr Walsh reported being employed by the NICHD outside the submitted work. No other disclosures were reported.

Funding/Support: The National Institutes of Health and the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) (U10 HD27871, U10 HD53119, UG1 HD21364, UG1 HD21373, UG1 HD21385, UG1 HD27851, UG1 HD27853, UG1 HD27856, UG1 HD27880, UG1 HD27904, UG1 HD34216, UG1 HD36790, UG1 HD40492, UG1 HD40689, UG1 HD53089, UG1 HD53109, UG1 HD68244, UG1 HD68270, UG1 HD68278, UG1 HD68263, UG1 HD68284, UG1 HD87226, UG1 HD87229) and the National Center for Advancing Translational Sciences (NCATS) (UL1 TR6, UL1 TR41, UL1 TR42, UL1 TR77, UL1 TR93, UL1 TR442, UL1 TR454, UL1 TR1117) provided grant support through cooperative agreements for the Neonatal Research Network.

Role of the Funder/Sponsor: NICHD staff provided input into the study design, conduct, analysis, and manuscript drafting; NCRR and NCATS cooperative agreements provided infrastructure support to the NRN. For the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) Neonatal Research Network (NRN), RTI International had full access to all the data in the study and, with the NRN center principal investigators, takes responsibility for the integrity of the data and accuracy of the data analysis. While NICHD staff had input into the study design, conduct, analysis, and manuscript drafting, the comments and views of the authors do not necessarily represent the views of NICHD, the National Institutes of Health (NIH), the Department of Health and Human Services, or the US Government. The clinical sites as well as the Data Coordinating Center (DCC) participating in this trial were all part of the NICHD NRN, a cooperative agreement grant established by NICHD, NIH, to improve health care and outcomes for newborns by conducting rigorous multicenter clinical trials in neonates. NRN centers and the DCC are chosen competitively every 5 to 7 years through open competition as per usual NIH peer review processes. Patient or public involvement in the design, conduct, and reporting of the trial are not applicable because the Community Engagement Board was not in operation at the time the BiB trial was designed and started. Participating NRN sites collected data and transmitted it to RTI International, the DCC for the network, which stored, managed, and analyzed the data for this study. Individual patient data from this trial, suitably deidentified, along with associated documentation will be available on the NICHD Data and Specimen Hub (N-DASH).

Disclaimer: While NICHD staff had input into the study design, conduct, analysis, and manuscript drafting, the comments and views of the authors do not necessarily represent the views of NICHD, the National Institutes of Health, the Department of Health and Human Services, or the US government.

Group Information: Members of the Eunice Kennedy Shriver National Institute of Child Health and Human Development Neonatal Research Network are listed in Supplement 5.

Data Sharing Statement: See Supplement 6.

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